

TRD — Agent-Generated Preference Data for DPO Fine-Tuning

System architecture, tech stack, data schema, training configuration, and evaluation harness requirements.

Project: Agent-Generated Preference Data for DPO Fine-Tuning of a Small Language Model
Prepared for: Dhruv Singhal — CSE, RV College of Engineering (RVCE), Bengaluru — USN: 1RV24CS077
Document version: 1.0 | September 2026

Budget constraint: Every component of this project is designed to run at \$0 cost — free-tier compute (Google Colab T4), free/open-weight models, and free public datasets only.

1. System Architecture

The pipeline is a linear chain with one branch point (agent-generated vs. human-labeled data feeding the same downstream trainer, for the ablation):



2. Technology Stack

Component	Choice	Rationale
Base model	Qwen2.5-1.5B-Instruct	Apache 2.0, strong small-model instruct quality
Fine-tuning method	QLoRA (4-bit) via Unsloth + PEFT	2-4x faster, ~60-70% less VRAM than vanilla HF training on a T4
DPO implementation	Hugging Face TRL — DPOTrainer	Reference implementation, actively maintained, integrates natively with PEFT adapters

Component	Choice	Rationale
Agent orchestration	Python + one free-tier LLM API (Groq or Gemini free tier) or a local Ollama model	No cost; local Ollama removes rate-limit risk entirely
Embeddings (reward model features)	sentence-transformers (e.g. all-MiniLM-L6-v2)	Free, CPU-runnable, standard baseline embedding model
Reward model	scikit-learn LogisticRegression or a 2-layer MLP in PyTorch	Lightweight, trains in seconds/minutes on CPU
Explainability	SHAP (KernelExplainer or LinearExplainer depending on model choice)	Consistent with prior SHAP-based project work
Compute	Google Colab free tier (T4, 16 GB VRAM, ~12 GB system RAM)	Zero cost; session length a few hours, subject to Google's usage limits
Storage / versioning	Hugging Face Hub (free public/private repos) + local JSONL	Free hosting for datasets, LoRA adapters, and model cards
Evaluation datasets	TruthfulQA, HaluEval (subset), HH-RLHF prompts	All free and public on Hugging Face Datasets

3. Data Schema

3.1 Preference pair (JSONL, one object per line)

```
{
  "prompt": "Explain why the sky is blue in one paragraph.",
  "chosen": "<full text of the response ranked higher>",
  "rejected": "<full text of the response ranked lower>",
  "meta": {
    "source": "agent_pipeline",          // or "hh_rlhf_human"
    "num_agents": 4,
    "critic_confidence": 0.82,          // critic's self-reported confidence, 0-1
    "devils_advocate_used": true,
    "rubric_scores": {
      "helpfulness_chosen": 4, "helpfulness_rejected": 2,
      "correctness_chosen": 5, "correctness_rejected": 3
    }
  }
}
```

3.2 Reward-model training row (derived, CSV/Parquet)

```
prompt_id, response_id, is_chosen, embedding_cosine_to_prompt,
response_length_tokens, hedging_word_ratio, named_entity_count,
cross_exam_survived (bool), label <- 1 if this response was "chosen"
```

4. DPO Training Configuration

Hyperparameter	Value
base_model	Qwen/Qwen2.5-1.5B-Instruct
load_in_4bit	true (QLoRA)

Hyperparameter	Value
lora_r	16
lora_alpha	32
lora_dropout	0.05
target_modules	q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj
learning_rate	5e-6 – 5e-5 (sweep; start at 1e-5)
dpo_beta	0.1 (KL-penalty strength; sweep 0.05-0.3)
per_device_train_batch_size	2-4 (T4 VRAM constrained)
gradient_accumulation_steps	4-8 (effective batch size ~16-32)
num_train_epochs	1-3 (DPO is prone to overfitting past ~2-3 epochs on small data)
max_prompt_length / max_length	512 / 1024 tokens
optimizer	adamw_8bit (bitsandbytes)
precision	bf16 if supported, else fp16

5. Evaluation Harness Requirements

- **Win-rate:** for each held-out prompt, generate a response from base model and DPO-tuned model; a separate, larger free-tier judge model picks a winner (blind, order-randomized to avoid position bias); report win/tie/loss percentages with a binomial confidence interval.
- **Benchmark accuracy:** run a fixed subset (e.g. 200 questions) of TruthfulQA and HaluEval through both models using the same automated scoring script; report accuracy deltas.
- **Training diagnostics:** log DPO reward margin (chosen logprob – rejected logprob) and approximate KL-to-base-model over training steps; flag divergence as an early-stopping signal.
- **Ablation table:** three rows — base model (no DPO), DPO on agent data, DPO on human data — across every metric above, in one summary table.

6. Non-Functional Requirements

Requirement	Detail
Cost	\$0 — every component must run on free tiers (Colab free T4, free API tiers or local Ollama, free public datasets)
Reproducibility	Fixed random seeds; pinned library versions in a requirements.txt; config stored as YAML/JSON alongside each run
Runtime budget	Full pipeline (data gen + 2 DPO runs + evaluation) should fit in ≤ 3-4 Colab sessions of ≤ 3 hours each
Storage budget	LoRA adapters only (a few hundred MB each) are checkpointed, not full merged model weights, to stay within free Hugging Face Hub storage limits
License compliance	Only Apache 2.0 / MIT / permissively-licensed models and datasets used, to keep any resulting publication unencumbered

References & Links

Hugging Face. **TRL — DPOTrainer Documentation**. https://huggingface.co/docs/trl/dpo_trainer — *reference implementation used for training*

Hugging Face. **PEFT Documentation**. <https://huggingface.co/docs/peft/index> — *LoRA adapter library*

Unsloth AI. **Unsloth — GitHub Repository & Docs**. <https://github.com/unslothai/unsloth> — *free-tier-optimized LoRA/QLoRA/DPO training*

Unsloth AI. **Unsloth Documentation**. <https://docs.unsloth.ai> — *setup and notebook guides*

Qwen Team, Alibaba. **Qwen2.5-1.5B-Instruct — Model Card**. <https://huggingface.co/Qwen/Qwen2.5-1.5B-Instruct> — *primary recommended base model*

Anthropic. **hh-rlhf Dataset (Hugging Face Datasets)**. <https://huggingface.co/datasets/Anthropic/hh-rlhf> — *human-labeled preference data used as reference set*

Hugging Face Datasets. **truthful_qa Dataset**. https://huggingface.co/datasets/truthful_qa — *evaluation dataset*

RUCAIBox. **HaluEval — GitHub Repository (data + code)**. <https://github.com/RUCAIBox/HaluEval> — *evaluation dataset and scoring scripts*

Reimers, N., Gurevych, I. / SBERT.net. **Sentence-Transformers Documentation**. <https://www.sbert.net> — *embedding model used for reward-model features*

Lundberg, S. et al. **SHAP — GitHub Repository**. <https://github.com/shap/shap> — *explainability library used for reward-model analysis*

Google. **Google Colaboratory — Free Tier**. <https://colab.research.google.com> — *free T4 GPU compute used for all training*

Groq. **GroqCloud — Free API Tier**. <https://console.groq.com> — *one option for the free-tier agent-debate LLM calls*

Ollama. **Ollama — Run LLMs Locally**. <https://ollama.com> — *zero-cost, rate-limit-free local alternative for the agent layer*

End of document.